BENJAMIN ANDERSON

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Electronic Engineer

PROJECTS

Cube
A handheld device comprising six OLED screens, one for each face of a cube. This project required the design of circuits for battery management and voltage regulation, 3D modelling of the frame, and efficient C code to run six screens and a six-DOF sensor from a single ESP32.
`KiCad / FreeCAD / Git / Blender`

Orthodox
A unique ortholinear keyboard. This project required rapid prototyping of custom keycaps using parametric designs, PCB coils around key switches to measure analogue key presses with an LDC chip, and a custom driver to allow seamless integration with computers.
`KiCad / FreeCAD / Git / OpenSCAD`

Avalanche
Modelled a near-infrared InGaAs/InAlAs avalanche photodiode in C++, using a random path length model to select layer widths against responsivity, noise, breakdown and capacitance specifications.
`C++ / MATLAB / LaTeX`

EXPERIENCE

Support Host — **Caledonian Sleeper** 2024-2026
- Managed passenger boarding, ticket checks and payments, while responding to service issues and complaints with professionalism and tact.
- Developed strong interpersonal and conflict-resolution skills when dealing with difficult customers, resolving concerns and communicating service disruptions and alternative travel arrangements.
- Coordinated time-critical morning operations, ensuring passengers were awake, prepared and ready to disembark on schedule, while liaising with station staff to arrange breakfast orders and food deliveries.
- Responded to safety and security incidents, including medical emergencies, fire procedures and unauthorised attempts to board the train, following established procedures and contacting emergency services when required.
- Maintained clear communication with the Train Manager and passengers during significant journey disruptions, helping coordinate practical solutions in challenging situations, including remote areas of Scotland.

RF Test Engineer — **Curvalux** 2021-2022
- Used LaTeX to produce clean, well-formatted reports from data and graphs generated with Matplotlib.
- Collaborated on and documented repositories using GitLab, Jira, Confluence, and Slack.
- Developed Python libraries to control rotators, switches, Keysight VNAs, and beamforming ICs, automating radiation pattern measurements in a custom anechoic chamber.
- Translated MATLAB algorithms for phase tapering and beam steering of a phased array antenna into working code on early prototype hardware.

Research & Development Engineer — **Huawei** 2019-2020
- Analysed device reliability using PAM4/NRZ inspection, shear testing, and electroluminescence inspection.
- Built a standalone program with PyQt to automatically produce failure analysis reports from data gathered by lab technicians, saving countless engineering hours on a repetitive manual task.
- Developed a Python library to communicate with a TEC controller over a serial protocol.
- Led weekly meetings to update senior engineers on device failures and reliability.

Hardware Engineer — **Avalon** 2018-2019
- Designed and soldered the power management PCB for an underwater ROV.
- Met with suppliers to discuss partnerships and sponsorships for the competition.
- Regularly liaised with engineers from different disciplines to check compatibility.
- Managed my time effectively to balance university commitments with those to Avalon.

VOLUNTEERING

Trail Maintenance & Stonework — **Stonewalling Ticino, Switzerland** — 1.5 months

- Built traditional loose stone walls and helped create new flat land on the hilly terrain in Lavizzara, designed to support vehicle access for the local area.
- Travelled by helicopter to work at altitude and cleared overgrown trails with a team using strimmers.

Restoration Volunteer — **Barnton Bunker, Edinburgh** — 2 months

- Carried out renovation work on a Cold War-era nuclear bunker, including cleaning, wall preparation, plastering, wall cladding, and installing electrical conduit.

Land & Animal Care — **Western Ghats, near Madikeri, India** — 1.5 months

- Maintained rural land, carried out household repairs, and managed the welfare of dogs and livestock.
- Learned Buddhist philosophy, meditation, and breathing techniques from an elderly Buddhist practitioner.

Temple Maintenance & Community Volunteering — **Kongoji Zen Temple, Minamiashigara, Japan** — 2 months

- Maintained temple grounds and Zen gardens through weeding, watering, and gravel raking, helping preserve the temple's environment.
- Took part in daily 5:00 am Zen meditation and morning temple rituals, including ringing the temple bell.
- Assisted with preparations for the Tanabata Star Festival, supporting the organisation of a traditional Japanese cultural event.

REFERENCES

Jason Hanson — **Principal RF Engineer**
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+44 (0)7799 036 362

Xiaoli Chu — **PhD in Wireless Communications**
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Lu Zhang — **Senior Reliability Engineer**
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EDUCATION

Bachelor of Engineering — **University of Sheffield** — 2017-2021
Electronic Engineering: 1st

A-levels — **Dunraven** — 2015-2017

Mathematics	B
Physics	B
Computing	B

GCSEs (11 qualifications passed) — **Harris Academy** — 2010-2015

Mathematics	A
Science	A
English	B